

CS101: Introduction to Computer Science

Academic Year 2025–26 | Semester 1 | Credits: 4

Course Description

This course introduces students to the fundamental concepts of computer science including algorithm design, data structures, object-oriented programming, and computational thinking. By the end of the course, students will be able to analyse algorithmic complexity, implement core data structures, and apply OOP principles in practical projects.

Syllabus Topics

The following 12 topics form the complete syllabus. All topics are examinable.

#	Topic	Schedule
1	Introduction to Algorithms & Complexity	Week 1–2
2	Arrays and Linked Lists	Week 2–3
3	Stacks and Queues	Week 3
4	Recursion and Backtracking	Week 4
5	Sorting Algorithms (Bubble, Merge, Quick)	Week 5
6	Binary Search and Divide & Conquer	Week 5–6
7	Trees and Binary Search Trees (BST)	Week 6–7
8	Graphs: BFS and DFS Traversal	Week 7–8
9	Dynamic Programming Fundamentals	Week 8–9
10	Hashing and Hash Tables	Week 9
11	Object-Oriented Programming Principles	Week 10–11
12	File Handling and Exception Management	Week 11–12

Assessment

- Mid-term Exam (Topics 1–6): 30%
- End-term Exam (Topics 1–12): 50%
- Assignments & Practicals: 20%